Software Requirements Specification

Smart Health and Fitness Tracker

1. Introduction

1.1 Purpose

It describes what the system must do and the qualities it must have.

1.2 Scope

smart health and fitness tracker is a standalone desktop application that lets the user record and understand their daily health and fitness habits. Each user has a private account and can log steps, exercise sessions, water intake, sleep, and body weight. The system turns these records into useful information: estimated calories burned, BMI, a 0 to 100 Daily Wellness Score, progress against personal targets and goals, weekly and monthly reports, and charts.

System goals:

- Make it quick to record daily health data.
- Give clear, transparent feedback on progress using simple, explainable rules.
- Keep each user's health data private and safe on their own computer.
- Motivate consistent healthy habits without giving medical advice.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
SHFT	Smart Health and Fitness Tracker, the system described here
FR / NFR	Functional Requirement / Non-Functional Requirement
UC	Use Case

Term	Meaning
UML	Unified Modeling Language
GUI / UI	(Graphical) User Interface
BMI	Body Mass Index = weight (kg) / height (m) squared
MET	Metabolic Equivalent of Task, a standard measure of exercise intensity
kcal	Kilocalorie, the unit used for calories burned
ml	Millilitre, the unit used for water intake
Daily Target	A per-day goal for steps, active minutes, water, or sleep
Fitness Goal	A longer-term goal with a target value and a deadline
Wellness Score	A 0 to 100 daily indicator computed from the four targets
Streak	Number of consecutive days meeting a condition
Badge	An achievement awarded once when a defined condition is met
SQLite	Lightweight file-based relational database
Tkinter	Python's standard GUI toolkit
PBKDF2	Password-based key derivation function used to hash passwords
CRUD	Create, Read, Update, Delete

1.4 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
2. ISO/IEC/IEEE 29148:2018, Systems and software engineering, Life cycle processes, Requirements engineering.
3. Dr. Mohamed Sami Rakha, CSAI 203 Course Project description, Fall 2026.

4. Ainsworth, B. E. et al., 2011 Compendium of Physical Activities: a second update of codes and MET values, Medicine & Science in Sports & Exercise, 43(8). (Source of the MET lookup values used in FR-06.)
5. World Health Organization, BMI classification for adults.
6. Python Software Foundation, Python 3 documentation (tkinter, sqlite3, hashlib, datetime, unittest). https://docs.python.org/3/
7. SQLite documentation. https://www.sqlite.org/docs.html
8. Matplotlib documentation. https://matplotlib.org/

1.5 Overview

this section gives the overall description: how the product fits its environment, its main functions, users, stakeholders, operating environment, constraints, documentation, and assumptions. Section 3 contains the specific requirements: detailed functional requirements (3.1), the use case model with diagrams and descriptions (3.2), the domain model with a conceptual class diagram and class descriptions (3.3), non-functional requirements with test methods (3.4), and external interface requirements (3.5). Section 4 contains appendices: a data dictionary of the main database tables and a glossary.

2. Overall Description

2.1 Product Perspective

SHFT is a new, self-contained product. It is not part of a larger suite and does not depend on any server. it is a stand alone app

2.2 Product Functions

Category	Functions
Account	Registration (FR-01), secure login and logout (FR-02), profile management (FR-03)
Tracking	Daily steps (FR-04), exercise sessions (FR-05), water intake (FR-07), sleep (FR-08), weight (FR-09)
Calculation	Calorie expenditure estimation (FR-06), BMI (FR-09)

Category	Functions
Goals	Customizable daily targets (FR-10), fitness goals with completion percentage (FR-11)
Insight	Daily wellness score (FR-12), daily dashboard (FR-13), weekly and monthly reports (FR-14), progress charts (FR-15)
Data management	Record history with filtering, editing, and deletion (FR-16)
Engagement	Custom reminders (FR-17), streaks and milestones (FR-18), achievement badges (FR-19), daily challenge and suggestions (FR-20)

2.3 User Classes and Characteristics

User class	Description	Experience level
Guest	A person who is not logged in (a new visitor or a returning user) and can only see the Login and Registration screens.	None required
Registered User	A person aged 13 to 100 who tracks personal health data. Has full access to all features, but only to their own data.	Low to moderate technical skill. Familiar with basic desktop programs. Needs no health expertise.
System Scheduler (internal actor)	The application's internal timer that checks reminders every 30 seconds while the app is open.	Not a human

The system has no administrator role: it is a local single-computer application, and each account's data is isolated from the others.

2.4 Software Stakeholders

Stakeholder	Interest in the software
End users (Registered Users)	Easy, accurate tracking, clear progress feedback, privacy of health data, motivation to stay consistent.
Development team	A clear, buildable, testable specification and a manageable scope within the semester.
Course instructor and teaching assistants	A complete, consistent SRS that follows the required outline and demonstrates software engineering practice.
Testers / QA	Measurable requirements and pass/fail criteria for every non-functional requirement.
Future maintainers	Modular, documented code that is easy to change.
Parents or guardians of younger users (13 to 17)	Assurance that data stays local and no medical advice is given.
Health professionals (indirect)	Assurance that the app does not present itself as a diagnostic or medical tool.

2.5 Operating Environment

- **Hardware:** typical student laptop or desktop, at least 4 GB RAM, 200 MB free disk space, display of at least 1280 x 720.
- **Operating system:** Windows 10 or 11, macOS, or Linux with a graphical desktop.
- **Runtime:** Python 3.10 or later with the standard library (tkinter, sqlite3, hashlib, datetime, unittest) plus Matplotlib.
- **Network:** none required. The application works fully offline.
- **Browsers:** not applicable, since SHFT is not a web application.

2.6 Design and Implementation Constraints

- Implemented in Python with Tkinter for the GUI and SQLite for storage; Matplotlib for charts.

- Passwords must be stored only as salted hashes (hashlib, for example PBKDF2 with a random salt). All SQL must be parameterized.
- No machine learning; challenges, suggestions, scores, and reports use transparent rules.
- Health outputs (BMI, wellness score, calories, challenges, medication reminders) must be labeled as general indicators and must not give medical advice.
- Personal health data must stay on the user's computer and not be transmitted.
- MET values must come from a published reference (see reference 4).
- The project must be completed within the semester schedule and limited team size.

2.7 User Documentation

The system will provide: a FAQ document covering registration, logging data, targets versus goals, how scores and calories are calculated, and troubleshooting; a short quick-start guide (README); in-application help such as the 'How is this calculated?' button for the wellness score, and tooltips and descriptive validation messages on forms.

2.8 Assumptions and Dependencies

Assumptions:

- Users enter their own data manually and honestly; the system cannot verify it.
- The application runs on one computer per user, and users do not edit the database file directly.
- The computer's system clock and date are correct, since streaks, reminders, and reports depend on them.
- Reminders are needed only while the application is open.

Dependencies:

- Python 3.10 or later and its standard library (tkinter, sqlite3, hashlib, datetime, shutil).
- Matplotlib for chart drawing.
- A published MET reference for the exercise lookup table.
- The local file system for the database, backups, logs, and exports.

3. Specific Requirements

3.1 Functional Requirements

FR-01: User Registration. The system shall let a new user create an account with a username, password, full name, and an optional email. Usernames must be unique, 4 to 20 characters, using letters, digits, and underscores only. Passwords must have at least 8 characters including at least one letter and one digit, must be re-entered for confirmation, and are never stored in plain text (salted and hashed). The system validates every field, rejects duplicate usernames, creates the account with default targets and an isolated set of records, and shows a success message with a redirect to Login. On error, no record is created and a message appears next to the offending field.

FR-02: Secure Login and Logout. The system shall authenticate users before granting access to any health data by hashing the entered password with the stored salt and comparing it to the stored hash. After 5 consecutive failed attempts for a username, login for that username is locked for 5 minutes, and the lockout message shows the remaining time. Failures show a generic message ('Invalid username or password'). A session lasts until the user logs out or closes the application, and a Logout button is available on every main screen.

FR-03: Profile Management. The system shall let a logged-in user view and update age (13 to 100), gender, height (50 to 250 cm), current weight (20 to 300 kg), and activity level (Low, Moderate, High). The user can also change the password by supplying the old one. On save the system validates the input, updates the profile, recalculates BMI, and adds a weight record for today if the weight changed. Errors name the invalid field and the allowed range.

FR-04: Daily Step Logging. The system shall let the user record the steps for a date (default today) as an integer from 0 to 100,000. Future dates are rejected. If a record already exists for that date, the system asks whether to replace it or add to it. After saving it shows an updated 'Steps today: X / target (Y%)' label.

FR-05: Exercise Session Logging. The system shall let the user record an exercise session by choosing a type from a predefined list (for example walking, running, cycling, swimming, yoga, strength training), a duration in minutes (1 to 600), an intensity (Light, Moderate, Vigorous), and an optional note. Several sessions per day are allowed. The system validates input, obtains the estimated calories from FR-06, stores the session, updates the day's active minutes, and shows a confirmation with the estimated calories.

FR-06: Calorie Expenditure Estimation. The system shall estimate calories burned using calories = MET x body weight (kg) x duration (hours), where MET comes from a fixed lookup table by exercise type and intensity, taken from the published reference in Section 1.4. The result is rounded to a whole number and labeled an approximate estimate. The user's latest weight is used; if none exists, the user is asked to complete the profile. A 'Calorie Calculator' screen lets the user test a hypothetical session without saving it.

FR-07: Water Intake Tracking. The system shall let the user log water in millilitres, either by typing 1 to 5,000 ml per entry or using quick-add buttons (for example 250 ml and 500 ml). Each entry is stored with a timestamp, the daily total is recomputed, and a progress bar and text such as '1,500 / 2,500 ml (60%)' are updated. If the daily total exceeds 5,000 ml, a gentle notice suggests checking the entry.

FR-08: Sleep Duration Logging. The system shall let the user record sleep for a date either as bed time and wake time (including bed times before midnight) or as total hours. The duration must be between 0.5 and 24 hours. The system classifies the night as Below target, On target, or Above target compared with the sleep target, and keeps one record per date (a repeated entry prompts for replacement). A message such as 'You slept 7.5 hours (On target)' is shown.

FR-09: Weight Tracking and BMI Calculation. The system shall let the user log weight (20 to 300 kg) with a date and keep the full history. It computes BMI = weight / height (m) squared and shows the adult category (Underweight below 18.5, Normal 18.5 to 24.9, Overweight 25 to 29.9, Obese 30 and above) with a note that BMI is a general indicator and not a diagnosis. It also shows weight change since the first entry. If height is missing, the user is asked to complete the profile.

FR-10: Customizable Daily Targets. The system shall let each user set daily targets for steps, active minutes, water (ml), and sleep (hours). Defaults (8,000 steps, 30 minutes, 2,000 ml, 8 hours) are provided at registration, can be changed at any time within valid ranges, and can be restored with 'Reset to Defaults'. Changed targets apply from the save date, so earlier records keep their original percentages. All progress calculations use these targets.

FR-11: Fitness Goal Management and Completion Percentage. The system shall let the user create longer-term goals with a type (reach a target weight, exercise N minutes per week, walk N total steps, or sleep N hours per night), a target value, and a deadline that must be in the future. The system calculates completion as min(current / target, 1) x 100 (weight goals use progress from the starting weight), recomputes it whenever a related record is saved, and sets the status to Active, Completed, or Expired. Goals can be edited or deleted, and a notice appears on completion.

FR-12: Daily Wellness Score. The system shall compute a score from 0 to 100 using fixed points: steps 30, active minutes 25, water 20, sleep 25. Each component earns points in proportion to its achieved percentage of target, capped at 100%; missing data counts as zero. The rounded total maps to a label: 85 to 100 Excellent, 60 to 84 Good, 30 to 59 Fair, 0 to 29 Needs Attention. A 'How is this calculated?' button shows the four-line breakdown with a disclaimer that the score is a general indicator, not a medical assessment.

FR-13: Daily Summary Dashboard. The system shall show a dashboard as the main screen after login. For the selected date (default today) it shows steps, active minutes, calories burned, water, sleep, latest weight and BMI, wellness score, active goals, streak, and the daily challenge. The user moves between dates with Previous/Next buttons or a date picker, and uses quick-action buttons to open logging screens. The dashboard refreshes whenever a record changes.

FR-14: Weekly and Monthly Progress Reports. The system shall generate on-screen reports for a chosen week or month containing totals and daily averages for steps, active minutes, calories, water, and sleep; best and worst day; average wellness score; days each target was met; weight change; and a comparison with the previous period (increase, decrease, unchanged). A short rule-based text summary is included, and the report can be saved as a text file. Periods without data show 'No data available'.

FR-15: Progress Charts. The system shall display a line chart for weight, bar charts for steps, water, and sleep, and a line chart for wellness score. The user selects the metric and a period of 7, 30, or 90 days. Target lines are drawn on relevant charts, bar charts fill missing days with zero, and axes and titles are labeled. With fewer than 2 data points a message is shown. The chart can be saved as a PNG image.

FR-16: Record History, Editing, and Deletion. The system shall provide a history view for steps, exercise, water, sleep, and weight records with filtering by type and date range, search of exercise sessions by type, and sorting by date. Users can edit a record (with the same validation as creation) or delete it after a confirmation prompt. Only the current user's records are affected. After a change, the dashboard, goals, score, streaks, and the calories of edited exercise sessions are recalculated.

FR-17: Custom Reminders. The system shall let users create reminders with a title, time of day, repeat pattern (Daily or selected weekdays), and category (Water, Exercise, Sleep, Medication, Other). Reminders can be enabled, disabled, edited, or deleted. While the application is open, a check every 30 seconds shows a popup at the due time once per scheduled time, with 'Done' and 'Snooze 10 minutes' options. Reminders missed while the app was closed are listed as 'Missed' at the next start. Medication reminders contain only user-entered text and give no medical advice.

FR-18: Streak Tracking and Progress Milestones. The system shall track a Consistency Streak (consecutive days with at least one chosen criterion met; default: at least 70% of the step target or at least 20 active minutes) and a Logging Streak (consecutive days with any record). Current and longest streaks are shown, with milestone notices at 3, 7, 14, 30, and 60 days. Streaks are recalculated after every saved, edited, or deleted record, and the user can choose the criterion.

FR-19: Achievement Badges. The system shall award badges when conditions are met, for example First Log, First Goal Completed, 7-Day Streak, Hydration Hero (water target met 5 days in a row), Sleep Champion (sleep target met 5 days in a row), 100,000 Total Steps, and Weight Goal Reached. Each badge is awarded once and stored with its date. An 'Achievement Unlocked' popup appears when earned, and a badge gallery shows earned badges and the unlock conditions of locked ones. Deleting records does not revoke badges.

FR-20: Personalized Daily Challenge and Activity Suggestions. The system shall create one daily challenge and up to three activity suggestions using simple rules. On first login each day it computes 7-day averages, finds the metric with the lowest percentage of target, and selects a matching challenge from a predefined list with a target slightly above the recent average. The user can Accept, Skip, or view suggestions; the status is Pending, Accepted, Completed, or Skipped, and becomes Completed when the user's data meets the condition. All text is general wellness guidance with a disclaimer.

3.2 Use Case Model

The use case model has three actors. The **Guest** is a person who is not logged in, either a new visitor or a returning user. The **Registered User** is a logged-in person who tracks their health. The **System Scheduler** is the internal timer that triggers reminders. A Guest becomes a Registered User after registering and logging in. The twenty use cases are grouped into four diagrams (Figures 1 to 4) so that each figure stays readable.

3.2.1 Use Case Diagram

Figure 1. Guests register and log in; users log out and manage profiles

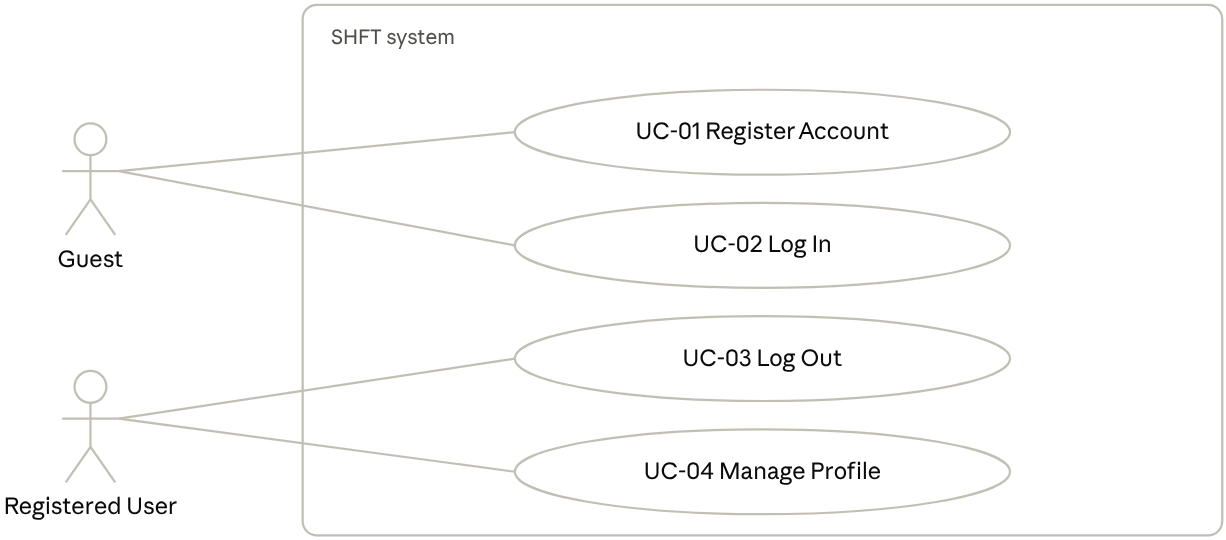


Figure 1 · account use cases · 4 use cases

Figure 1. Account use cases: UC-01 Register Account, UC-02 Log In, UC-03 Log Out, UC-04 Manage Profile.

Figure 2. One user logs five kinds of health data

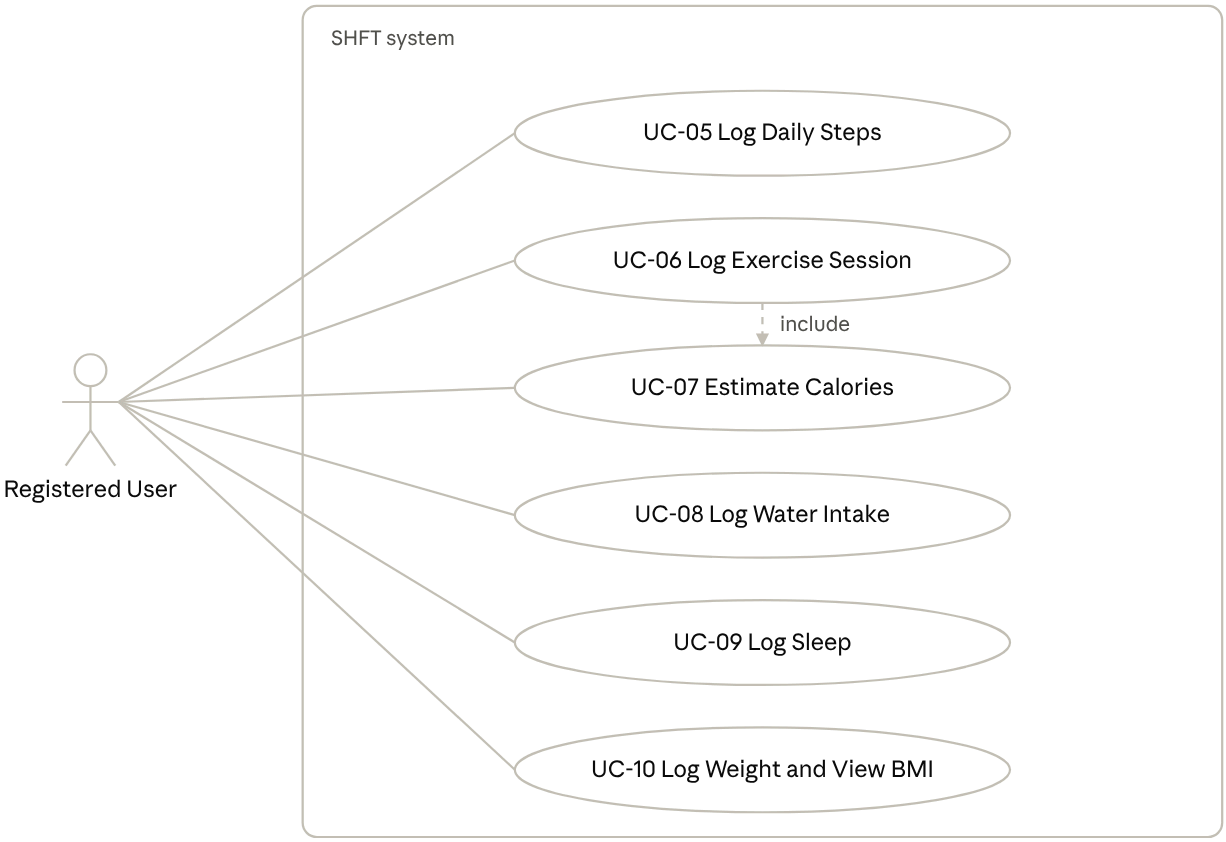


Figure 2 · tracking use cases · 6 use cases

Figure 2. Tracking use cases: UC-05 to UC-10. Logging an exercise session includes the calorie estimate.

Figure 3. Targets, goals, and insight views built on logged data

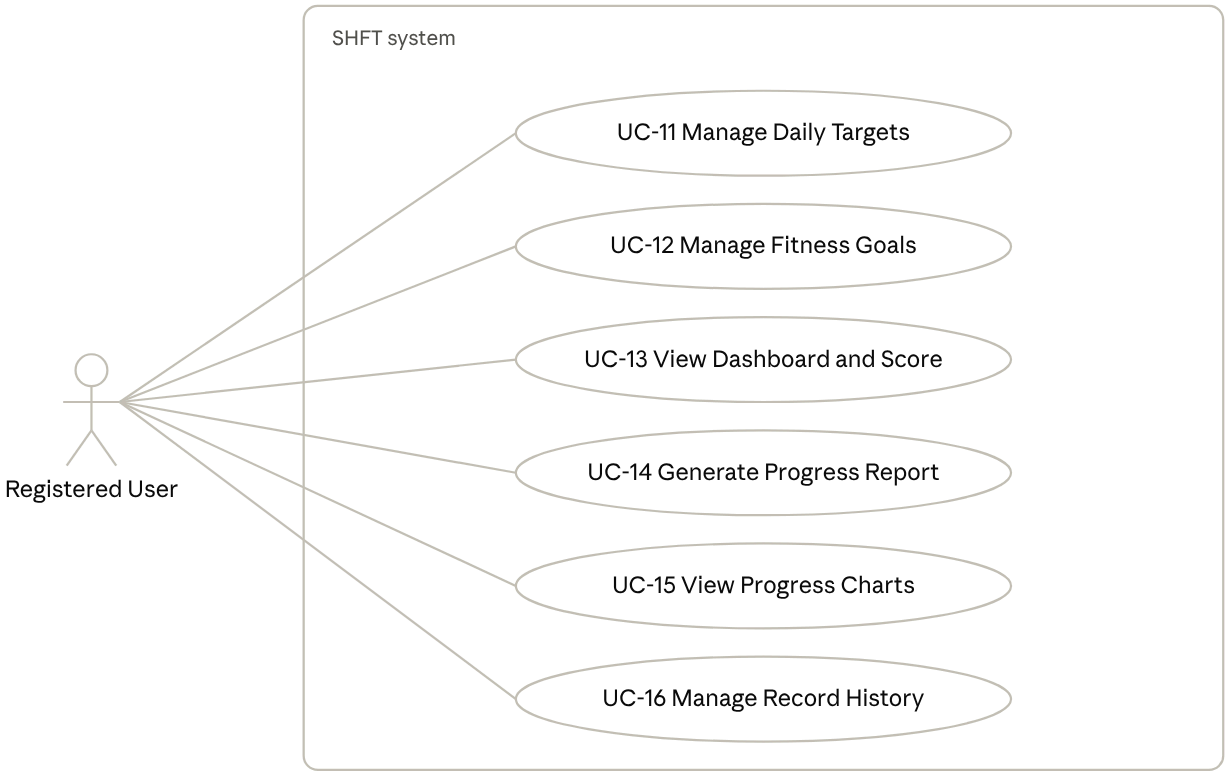


Figure 3 · targets, goals, insight · 6 use cases

Figure 3. Targets, goals, and insight use cases: UC-11 to UC-16.

Figure 4. Reminders are triggered by the scheduler; the rest by the user

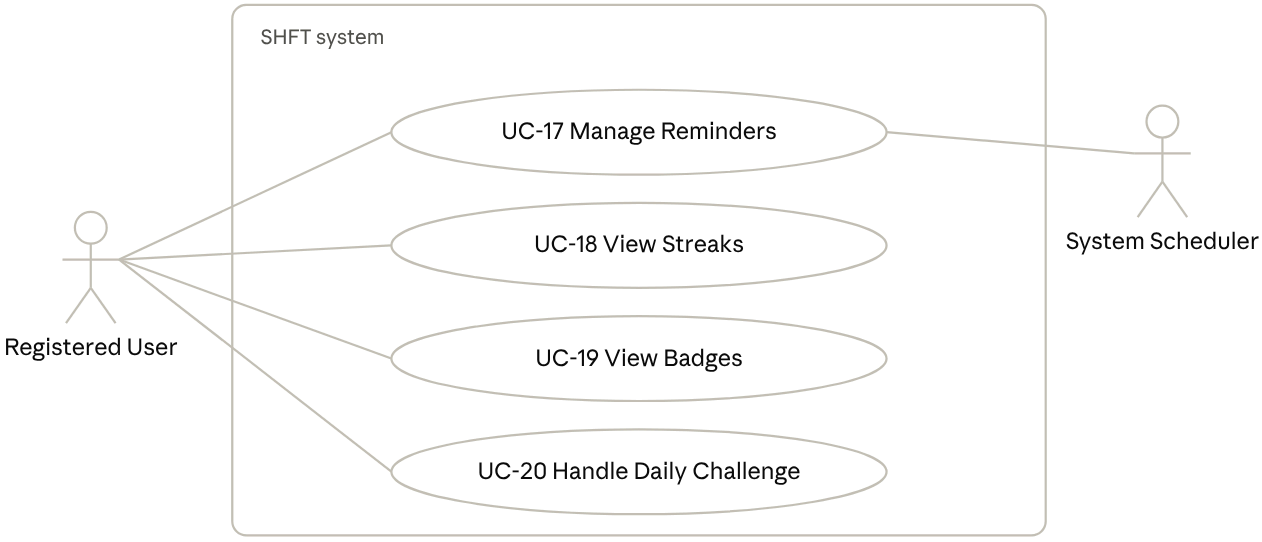


Figure 4 · engagement use cases · 4 use cases

Figure 4. Engagement use cases: UC-17 to UC-20. The System Scheduler triggers reminder popups.

3.2.2 Use Case Descriptions

Each use case is described once, using the IEEE-style template: ID, name, actors, related requirement, description, preconditions, trigger, main flow, alternative flows, postconditions, and business rules. In all use cases, 'the user' means a Registered User unless stated otherwise.

UC-01: Register Account

Field	Description
Actor	Guest
Related requirement	FR-01
Description	A new person creates an account to use the system.
Preconditions	The application is running and nobody is logged in.
Trigger	The Guest clicks 'Create Account' on the Registration screen.
Main flow	(1) Guest opens the Registration screen. (2) Guest enters username, password, password confirmation, full name, and optional email. (3) Guest clicks 'Create Account'. (4) System validates all fields and checks that the username is unused. (5) System salts and hashes the password, creates the account with default targets, and stores it. (6) System shows a success message and opens the Login screen.
Alternative flows	4a. A field is invalid: the system shows a message next to the field and creates nothing. 4b. The username exists: 'Username already exists'. 4c. The passwords differ: 'Passwords do not match'.
Postconditions	A new account with its own empty set of records and default targets exists. The Guest is not yet logged in.

UC-02: Log In

Field	Description
Actor	Guest
Related requirement	FR-02

Field	Description
Description	A person with an account authenticates to access their data.
Preconditions	The account exists and is not locked.
Trigger	The Guest clicks 'Login'.
Main flow	(1) Guest enters username and password. (2) System loads the stored salt and hash, hashes the entered password, and compares. (3) System resets the failed-attempt counter and starts the session. (4) System shows the dashboard (UC-13).
Alternative flows	2a. Credentials are wrong: 'Invalid username or password', and the failed-attempt counter increases. 2b. This is the 5th consecutive failure: the username is locked for 5 minutes. 2c. The account is locked: the system shows the remaining lock time and rejects the attempt even if the password is correct.
Postconditions	A session is active for the user, who becomes a Registered User.

UC-03: Log Out

Field	Description
Actor	Registered User
Related requirement	FR-02
Description	The user ends the current session.
Preconditions	The user is logged in.
Trigger	The user clicks 'Logout' on any main screen.
Main flow	(1) User clicks 'Logout'. (2) System clears the session. (3) System shows the Login screen.
Alternative flows	None. Closing the application also ends the session.

Field	Description
Postconditions	No user data is accessible until the next login.

UC-04: Manage Profile

Field	Description
Actor	Registered User
Related requirement	FR-03
Description	The user views and updates personal details and changes the password.
Preconditions	The user is logged in.
Trigger	The user opens 'My Profile'.
Main flow	(1) System shows current age, gender, height, weight, and activity level. (2) User edits the fields and clicks 'Save'. (3) System validates the values, updates the profile, and recalculates BMI. (4) If weight changed, the system adds a weight record for today. (5) System shows a confirmation.
Alternative flows	3a. A value is out of range: the system names the field and the allowed range and saves nothing. 2a. Change Password: the user enters the old and new passwords; the system verifies the old one, validates the new one, and stores the new hash. If the old password is wrong, an error is shown.
Postconditions	The profile (and password, if changed) is updated.

UC-05: Log Daily Steps

Field	Description
Actor	Registered User
Related requirement	FR-04
Description	The user records the number of steps for a date.

Field	Description
Preconditions	The user is logged in.
Trigger	The user opens the steps logging screen.
Main flow	(1) User selects a date (default today) and types the step count. (2) User clicks 'Save Steps'. (3) System validates the date and the integer from 0 to 100,000. (4) System stores the record and updates step progress. (5) System shows 'Steps today: X / target (Y%)'.
Alternative flows	3a. The date is in the future or the number is invalid: an error is shown and nothing is saved. 4a. A record already exists for the date: the system asks whether to replace it, add to it, or cancel.
Postconditions	The date's step total is stored; dashboard, goals, score, streaks, and badges are updated.

UC-06: Log Exercise Session

Field	Description
Actor	Registered User
Related requirement	FR-05
Description	The user records an exercise session with its estimated calories.
Preconditions	The user is logged in and has a weight on file.
Trigger	The user opens 'Log Exercise'.
Main flow	(1) User chooses the exercise type, duration (1 to 600 minutes), intensity, and an optional note. (2) User clicks 'Save Session'. (3) System validates the input. (4) System includes UC-07 to estimate calories. (5) System stores the session and updates active minutes for the day. (6) System shows the estimated calories and the updated 'Active minutes today'.

Field	Description
Alternative flows	3a. Input is invalid: an error naming the field is shown. 4a. No weight is on file: the system asks the user to complete the profile and does not save the session.
Postconditions	The session is stored with its estimated calories. Several sessions per day are allowed.

UC-07: Estimate Calories

Field	Description
Actor	Registered User
Related requirement	FR-06
Description	The system estimates calories burned; this use case is included by UC-06 and is also available as a stand-alone 'Calorie Calculator'.
Preconditions	The user is logged in.
Trigger	UC-06 saves a session, or the user opens the Calorie Calculator.
Main flow	(1) System receives exercise type, intensity, and duration. (2) System looks up the MET value. (3) System reads the user's latest weight. (4) System computes MET x weight (kg) x duration (hours) and rounds to a whole number. (5) System shows 'Estimated calories burned: N kcal (approximate)'.
Alternative flows	3a. No weight on file: the system asks the user to complete the profile.
Postconditions	In calculator mode nothing is stored. In UC-06 the result is stored with the session.

UC-08: Log Water Intake

Field	Description
Actor	Registered User
Related requirement	FR-07
Description	The user records water consumed.
Preconditions	The user is logged in.
Trigger	The user clicks a quick-add button or 'Add Water'.
Main flow	(1) User clicks a quick-add button (for example 250 ml or 500 ml) or types an amount. (2) System validates 1 to 5,000 ml. (3) System stores the entry with a timestamp and recomputes the day total. (4) System updates the progress bar and text such as '1,500 / 2,500 ml (60%)'.
Alternative flows	2a. The amount is invalid: an error is shown. 3a. The daily total exceeds 5,000 ml: a gentle notice asks the user to check the entry.
Postconditions	The entry is stored and the daily total is updated.

UC-09: Log Sleep

Field	Description
Actor	Registered User
Related requirement	FR-08
Description	The user records a night's sleep.
Preconditions	The user is logged in.
Trigger	The user opens the sleep logging screen.

Field	Description
Main flow	(1) User selects a date and enters bed time and wake time, or total hours. (2) User clicks 'Save Sleep'. (3) System computes the duration (including bed times before midnight) and validates 0.5 to 24 hours. (4) System saves it and classifies it as Below target, On target, or Above target. (5) System shows 'You slept 7.5 hours (On target)'.
Alternative flows	3a. The duration is invalid: an error is shown. 4a. A record exists for the date: the system asks whether to replace it.
Postconditions	One sleep record per date is stored.

UC-10: Log Weight and View BMI

Field	Description
Actor	Registered User
Related requirement	FR-09
Description	The user records body weight and sees BMI and weight change.
Preconditions	The user is logged in and the profile contains a height.
Trigger	The user opens the weight screen.
Main flow	(1) User enters weight (20 to 300 kg) and date and clicks 'Save Weight'. (2) System validates and stores it. (3) System computes BMI from the profile height and finds the category. (4) System shows 'BMI: 23.4 (Normal range). General indicator only.' and the weight change since the first entry.
Alternative flows	2a. The weight is invalid: an error is shown. 3a. Height is missing: the system asks the user to complete the profile.

Field	Description
Postconditions	The weight record is stored and the BMI panel is updated.

UC-11: Manage Daily Targets

Field	Description
Actor	Registered User
Related requirement	FR-10
Description	The user views and changes daily targets for steps, active minutes, water, and sleep.
Preconditions	The user is logged in.
Trigger	The user opens 'My Targets'.
Main flow	(1) System shows the four current targets. (2) User edits values and clicks 'Save'. (3) System validates each value and saves them. (4) System recalculates today's progress and updates the dashboard labels.
Alternative flows	3a. A value is invalid: an error names the field and range. 2a. The user clicks 'Reset to Defaults' (8,000 steps, 30 minutes, 2,000 ml, 8 hours).
Postconditions	New targets apply from the save date; earlier records keep their original percentages.

UC-12: Manage Fitness Goals

Field	Description
Actor	Registered User
Related requirement	FR-11
Description	The user creates, edits, deletes, and monitors longer-term fitness goals.

Field	Description
Preconditions	The user is logged in.
Trigger	The user opens 'Goals'.
Main flow	(1) System lists goals with completion percentage, remaining days, and status. (2) User clicks 'New Goal' and enters type, target value, and a future deadline. (3) System validates and stores the goal. (4) System recomputes progress whenever a related record is saved. (5) At 100% the status becomes Completed and a notice is shown.
Alternative flows	3a. The deadline is not in the future or the target is invalid: an error is shown. 2a. The user edits or deletes a goal after confirmation. 4a. The deadline passes before completion: the status becomes Expired.
Postconditions	The goal list reflects the latest data.

UC-13: View Dashboard and Score

Field	Description
Actor	Registered User
Related requirement	FR-12, FR-13
Description	The user sees a daily summary including the Wellness Score.
Preconditions	The user is logged in.
Trigger	Successful login, or a click on the dashboard in the menu.

Field	Description
Main flow	(1) System gathers the day's data and targets. (2) System shows steps, active minutes, calories, water, sleep, latest weight and BMI, goals, streak, and the challenge. (3) System computes and shows 'Wellness Score: N / 100 (label)'. (4) User may switch dates or click 'How is this calculated?' to see the four component points and the disclaimer.
Alternative flows	1a. No data exists for the date: the missing metrics count as zero.
Postconditions	None. The dashboard refreshes after any record change.

UC-14: Generate Progress Report

Field	Description
Actor	Registered User
Related requirement	FR-14
Description	The user generates a weekly or monthly report.
Preconditions	The user is logged in.
Trigger	The user opens 'Reports'.
Main flow	(1) User chooses Weekly or Monthly and a period and clicks 'Generate'. (2) System aggregates the records in the range and compares with the previous period. (3) System shows the summary table and a rule-based text summary. (4) Optionally the user clicks 'Save as Text File'.
Alternative flows	2a. The period has no data: 'No data available'. 4a. The file cannot be written: an error is shown and the on-screen report remains.
Postconditions	An optional text file exists.

UC-15: View Progress Charts

Field	Description
Actor	Registered User
Related requirement	FR-15
Description	The user views charts of progress over time.
Preconditions	The user is logged in.
Trigger	The user opens 'Charts'.
Main flow	(1) User selects a metric (weight, steps, water, sleep, or wellness score) and 7, 30, or 90 days and clicks 'Show Chart'. (2) System queries the range and draws the chart with labelled axes, a title, and target lines. (3) Optionally the user saves it as a PNG image.
Alternative flows	2a. Fewer than 2 data points: the system shows a message instead of a chart.
Postconditions	An optional PNG file exists.

UC-16: Manage Record History

Field	Description
Actor	Registered User
Related requirement	FR-16
Description	The user browses, filters, edits, and deletes past records.
Preconditions	The user is logged in and has records.
Trigger	The user opens 'History'.

Field	Description
Main flow	(1) User filters by record type and date range, searches exercise by type, and sorts by date. (2) User selects a row and clicks 'Edit' or 'Delete'. (3) For edit, the system validates the new values with the same rules as creation and saves them. (4) For delete, the system asks for confirmation and removes the record. (5) System recalculates dashboard values, goals, score, streaks, and exercise calories.
Alternative flows	3a. The edit is invalid: an error is shown and the record is unchanged. 4a. The user cancels the deletion: nothing changes.
Postconditions	Only the current user's records are changed. Earned badges are not revoked.

UC-17: Manage Reminders

Field	Description
Actors	Registered User (primary), System Scheduler (triggers popups)
Related requirement	FR-17
Description	The user creates and manages reminders and responds to them when due.
Preconditions	The user is logged in. For popups, the application must be open.
Trigger	The user opens 'Reminders', or the Scheduler's 30-second check finds a due reminder.

Field	Description
Main flow	(1) User clicks 'New Reminder' and enters title, time, repeat pattern, and category. (2) System validates and stores it. (3) User can enable, disable, edit, or delete reminders. (4) At the due time the Scheduler shows a popup with the title, once per scheduled time. (5) User clicks 'Done' or 'Snooze 10 minutes'.
Alternative flows	2a. A field is invalid: an error is shown. 4a. The reminder was due while the app was closed: it appears as 'Missed' at the next start.
Postconditions	Reminders are stored. A Medication reminder shows only the user's own text and no advice.

UC-18: View Streaks

Field	Description
Actor	Registered User
Related requirement	FR-18
Description	The user sees consistency and logging streaks and milestones.
Preconditions	The user is logged in.
Trigger	The user clicks 'Streaks' or views the dashboard.
Main flow	(1) System walks backward from today through consecutive qualifying days. (2) System shows 'Current streak: N days. Longest: M days' for both streak types. (3) User may change the streak criterion. (4) System announces a milestone at 3, 7, 14, 30, or 60 days.
Alternative flows	1a. Today has no qualifying data: the streak breaks.
Postconditions	Streaks are recalculated after any record is added, edited, or deleted.

UC-19: View Badges

Field	Description
Actor	Registered User
Related requirement	FR-19
Description	The user views earned and locked achievement badges.
Preconditions	The user is logged in.
Trigger	The user opens 'Badges', or a badge is earned after a save.
Main flow	(1) After a relevant save, the system evaluates the badge rules. (2) For each newly met rule it stores the badge with the date and shows an 'Achievement Unlocked' popup. (3) The Badges screen shows earned badges with dates and locked badges with their unlock conditions.
Alternative flows	2a. A badge is already earned: it is not awarded again.
Postconditions	Each badge is stored once and is never revoked.

UC-20: Handle Daily Challenge

Field	Description
Actor	Registered User
Related requirement	FR-20
Description	The user receives a daily challenge and suggestions and responds to them.
Preconditions	The user is logged in.
Trigger	First login of the day, or the user clicks 'Show suggestions'.

Field	Description
Main flow	(1) System computes 7-day averages and finds the metric with the lowest percentage of target. (2) System selects a matching challenge and shows 'Today's Challenge' with status Pending. (3) User clicks 'Accept' or 'Skip'. (4) User may click 'Show suggestions' to see up to three activity suggestions. (5) When the user's data meets the condition, the status becomes Completed.
Alternative flows	1a. Little or no history exists: the system selects a default starter challenge.
Postconditions	The challenge and its status are stored. All text is general wellness guidance with a disclaimer.

3.3 Domain Model

The domain model describes the main concepts the system stores and how they relate. A User owns all health data. Daily totals such as the Wellness Score, streaks, and reports are calculated on demand from these classes and are not stored as separate classes.

3.3.1 Conceptual Class Diagram

Figure 5. Every health record belongs to exactly one user

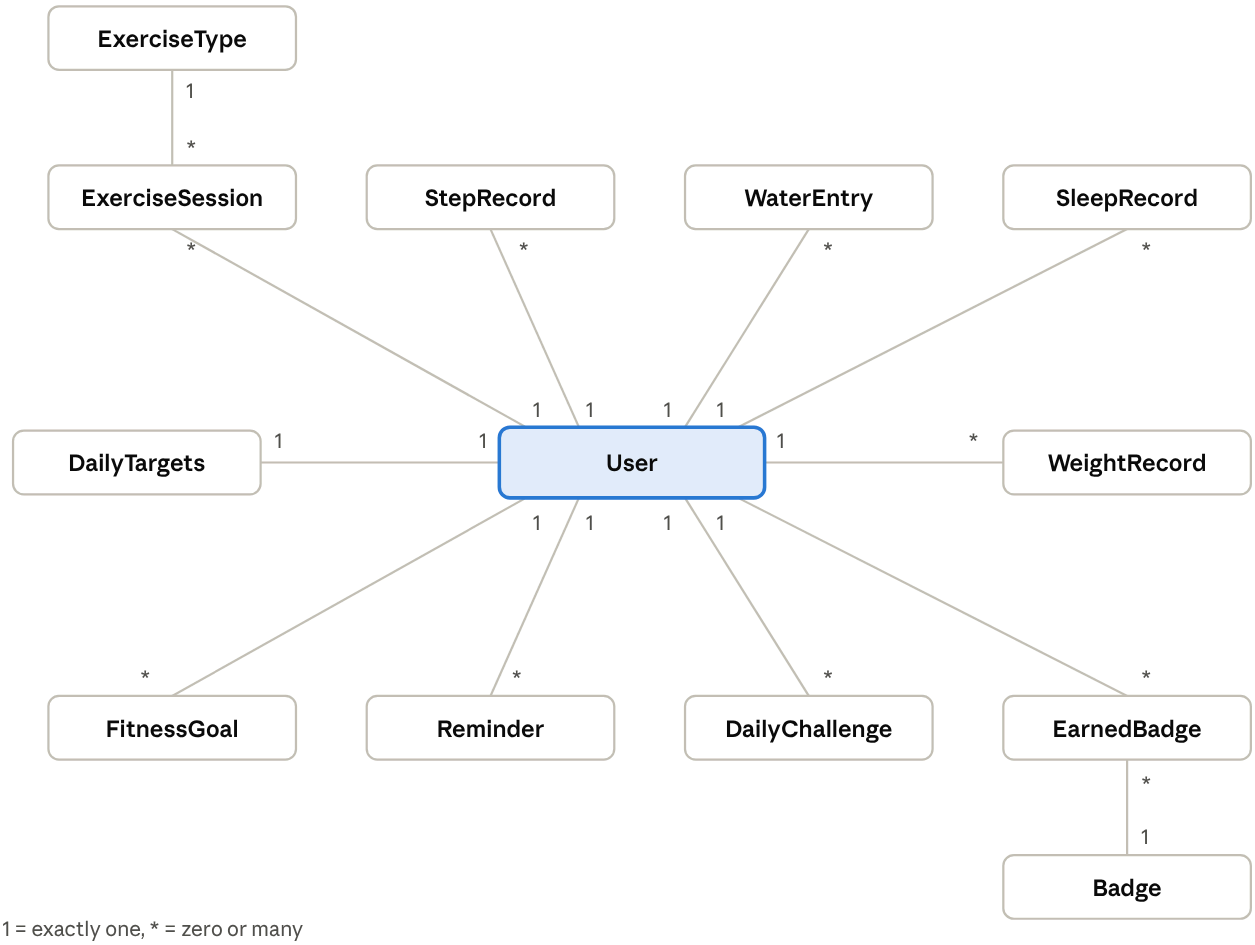


Figure 5 · conceptual class diagram · 13 classes

Figure 5. Conceptual class diagram. A User has exactly one DailyTargets object and many step, exercise, water, sleep, and weight records, goals, reminders, challenges, and earned badges. Each ExerciseSession refers to one ExerciseType, and each EarnedBadge refers to one Badge.

3.3.2 Class Descriptions

Class	Attributes	Associations	Purpose
User	userId, username, passwordHash, salt, fullName, email, age, gender, heightCm, weightKg, activityLevel, bmi, failedAttempts, lockUntil, streakCriterion	Owns one DailyTargets. Has many StepRecord, ExerciseSession, WaterEntry, SleepRecord, WeightRecord, FitnessGoal, Reminder, DailyChallenge, and EarnedBadge.	Represents an account holder and the profile data used in calculations. All other data is owned through this class.
DailyTargets	targetId, userId, stepsTarget, activeMinutesTarget, waterTargetMl, sleepTargetHours, effectiveFrom	Belongs to exactly one User (one-to-one).	Stores the personal daily targets used for progress percentages and the Wellness Score.
StepRecord	stepId, userId, date, stepCount	Belongs to one User. A User has at most one per date.	Stores the steps taken on a date.
ExerciseType	typeId, name, lightMet, moderateMet, vigorousMet	Has many ExerciseSession.	Predefined exercise types with the MET value for each intensity, used for calorie estimation.
ExerciseSession	sessionId, userId, typeId, date, durationMinutes, intensity, note, caloriesBurned	Belongs to one User and one ExerciseType.	Stores a single workout and its estimated calories. A day can have several.

Class	Attributes	Associations	Purpose
WaterEntry	waterId, userId, date, timestamp, amountMl	Belongs to one User.	Stores one water intake entry. The day's total is the sum of entries.
SleepRecord	sleepId, userId, date, bedTime, wakeTime, totalHours, status	Belongs to one User. A User has at most one per date.	Stores a night's sleep and its classification against the sleep target.
WeightRecord	weightId, userId, date, weightKg, bmi, bmiCategory	Belongs to one User.	Stores the weight history, with the BMI and category at that time.
FitnessGoal	goalId, userId, goalType, targetValue, startValue, deadline, completionPercentage, status	Belongs to one User.	Stores a longer-term goal and its progress, with status Active, Completed, or Expired.
Reminder	reminderId, userId, title, timeOfDay, repeatDays, category, isEnabled, lastTriggered, snoozedUntil	Belongs to one User.	Stores a user-defined reminder that is shown as a popup while the application is open.
Badge	badgeId, name, description, conditionKey	Has many EarnedBadge.	Defines an achievement and its unlock condition.
EarnedBadge	earnedId, userId, badgeId, dateEarned	Belongs to one User and one Badge.	Records that a user earned a badge on a date. Each badge is earned once per user.

Class	Attributes	Associations	Purpose
DailyChallenge	challengeId, userId, date, text, metric, targetValue, status	Belongs to one User.	Stores the day's challenge and its status: Pending, Accepted, Completed, or Skipped.

3.4 Non-Functional Requirements

The success criteria below are proposed and need team approval. Each requirement passes only if every listed criterion is met.

NFR-01: Performance (Responsiveness). The system shall respond quickly to everyday operations on a typical student laptop, even after months of use.

- *How it is tested:* seed the database with 1 user and 3 years of daily records (about 5,000 records across all tables). Measure elapsed time with Python's time.perf_counter() around each operation, running each 10 times on a computer with at least 4 GB RAM and no other heavy applications open.
- *Success criteria (averages):* saving a record (FR-04 to FR-09) 1 second or less; dashboard load (FR-13) 2 seconds or less; monthly report (FR-14) 3 seconds or less; 90-day chart (FR-15) 3 seconds or less; application start to login screen 5 seconds or less. No single run may exceed twice its limit.

NFR-02: Security (Account and Data Protection). The system shall protect accounts and personal health records from unauthorized access, including from other users of the same application.

- *How it is tested:* (1) inspect the SQLite file with a database viewer to see how passwords are stored; (2) enter SQL injection strings (for example ' OR '1'='1) in the login and text fields; (3) create two users and try to view or change one user's records while logged in as the other; (4) make 5 failed login attempts and check the lockout.
- *Success criteria:* 0 plain-text passwords, every password stored as a salted hash with a unique salt per user; 100% of injection strings fail to log in or alter data; 0 records of one user retrievable by another on any screen; the account is locked for 5 minutes after the 5th consecutive failure, and a 6th attempt during the lockout is rejected even with the correct password.

NFR-03: Usability (Ease of Use and Clarity). The system shall be easy for users with little technical experience to learn and use, with clear navigation, consistent screens, and understandable messages.

- *How it is tested:* a task-based test with 5 participants who have not seen the application. Each performs 6 tasks without instructions (register and log in, log steps, log a water entry, set a daily target, view the weekly report, delete a record) while an observer records completion and time. Each participant then rates ease of use from 1 to 5. The team also reviews all messages against a checklist.
- *Success criteria:* at least 80% of task attempts completed without help; a new user completes registration and the first record within 3 minutes; every main function reachable within 3 clicks from the dashboard; average rating 4.0 out of 5 or higher; 100% of validation errors name the problem field and the allowed range.

NFR-04: Reliability (Data Integrity and Recovery). The system shall store records correctly, avoid corrupting or losing data when errors or unexpected closings occur, and recover gracefully from invalid input or failures.

- *How it is tested:* (1) save 100 records through the interface and compare the database with the input; (2) force-close the application during 20 save operations and check for partial or corrupt records; (3) enter invalid input in all forms (empty, letters in numeric fields, out-of-range values) and watch for crashes; (4) delete the database file and start the application to test the backup restore prompt.
- *Success criteria:* 100% of the 100 saved records match the input; 0 corrupted or half-written records after 20 forced closings (each save is either fully stored or fully absent); 0 crashes or unhandled exceptions during invalid input tests; an automatic database backup at every startup with the last 3 kept, and a restore from the latest backup recovers all records.

NFR-05: Maintainability (Modular and Testable Code). The Python code shall be organized into separate, documented modules so team members can understand, change, and debug features without breaking others.

- *How it is tested:* code review against a checklist plus automated checks: count modules and function sizes, check docstrings, run unit tests with Python's unittest module, and count the calculation functions that have tests. A new team member is asked to change one default target value and one MET value, and the time is recorded.
- *Success criteria:* at least 5 modules with separate responsibilities (database, authentication, calculations, user interface, charts and reports); 100% of public functions and classes have docstrings; no function longer than 50 lines; every

calculation function (BMI, calories, wellness score, streaks, goal percentage) has at least 3 passing unit tests (normal, boundary, invalid); both change tasks done in 15 minutes or less with no other file changed.

3.5 External Interface Requirements

3.5.1 User Interface

The application has a single main window with a persistent navigation menu and consistent screens. Dates default to today and every form shows labelled fields, so no screen needs instructions. Errors and confirmations appear as message boxes or inline messages.

- **Registration and Login screens:** fields for username, password, confirmation, full name, and email; 'Create Account' and 'Login' buttons; lockout and error messages.
- **Dashboard:** date switcher (Previous, Next, date picker), metric cards with progress bars for steps, active minutes, calories, water, and sleep, a weight and BMI panel, the Wellness Score with a 'How is this calculated?' button, active goals, streak, 'Today's Challenge' card, and quick-action buttons.
- **Logging screens:** forms for steps, exercise (type list, minutes, intensity, note), water (quick-add 250 ml and 500 ml buttons plus custom amount), sleep (times or hours), and weight, plus the Calorie Calculator.
- **My Profile and My Targets:** pre-filled fields, 'Save', 'Change Password', and 'Reset to Defaults'.
- **Goals, History, Reminders, Badges, and Streaks screens:** lists or tables (goals with progress bars, a filterable history table with Edit and Delete, reminders with enable switches, a badge gallery with locked badges showing conditions).
- **Reports and Charts screens:** period and metric selectors, 'Generate' and 'Show Chart' buttons, and save buttons for the text report and PNG chart.
- **Popups:** reminder popup (Done, Snooze 10 minutes), milestone notices, and 'Achievement Unlocked'.

3.5.2 Hardware Interface

The system has no special hardware interfaces. It uses the computer's keyboard and mouse (or touchpad) for input and the display for output. It does not connect to printers, scanners, wearable devices, or sensors. Step counts and all other data are entered manually.

3.5.3 Software Interface

- **Operating system:** reads and writes local files (the SQLite database, backups, error log, text reports, PNG charts) through the Python standard library.
- **SQLite database:** accessed through Python's sqlite3 module using parameterized queries and transactions.
- **Tkinter:** provides all windows, widgets, dialogs, and the timer (after) used for reminders.
- **Matplotlib:** draws charts, embedded in Tkinter and exported to PNG.
- **Python standard library:** hashlib and os (password hashing), datetime (dates and times), shutil (backups), re (input checks), and unittest (tests).

3.5.4 Communication Interface

The application runs fully offline and makes no network calls, so no communication protocol is required in this version. All data stays on the user's computer. If a later version adds cloud backup or account sync, it must use HTTPS (TLS) for all communication.

4. Appendices

4.1 Appendix A: Data Dictionary

The following SQLite tables are planned. Every table that holds user data has a user_id column, and every query is filtered by it.

Table	Description	Main columns
users	Accounts, profile, and login protection	user_id, username, password_hash, salt, full_name, email, age, gender, height_cm, weight_kg, activity_level, bmi, failed_attempts, lock_until, streak_criterion
targets	One row of daily targets per user	target_id, user_id, steps_target, active_minutes_target, water_target_ml, sleep_target_hours, effective_from
steps	Daily step totals	step_id, user_id, date, step_count
exercise_types	Predefined exercise types and MET values	type_id, name, light_met, moderate_met, vigorous_met

Table	Description	Main columns
exercises	Exercise sessions	session_id, user_id, type_id, date, duration_minutes, intensity, note, calories_burned
water	Water entries	water_id, user_id, date, timestamp, amount_ml
sleep	Nightly sleep records	sleep_id, user_id, date, bed_time, wake_time, total_hours, status
weights	Weight history	weight_id, user_id, date, weight_kg, bmi, bmi_category
goals	Longer-term fitness goals	goal_id, user_id, goal_type, target_value, start_value, deadline, completion_percentage, status
reminders	User reminders	reminder_id, user_id, title, time_of_day, repeat_days, category, is_enabled, last_triggered, snoozed_until
badges	Badge definitions	badge_id, name, description, condition_key
user_badges	Badges earned by users	earned_id, user_id, badge_id, date_earned
challenges	Daily challenges	challenge_id, user_id, date, text, metric, target_value, status

4.2 Appendix B: Glossary

Term	Definition
Achievement Badge	A reward shown in the badge gallery, earned once when a defined condition is met.
Active minutes	Total minutes of logged exercise sessions in a day.
BMI	Body Mass Index: weight in kg divided by height in metres squared. A general indicator, not a diagnosis.

Term	Definition
Consistency Streak	Consecutive days on which the chosen streak criterion was met.
Daily Challenge	A rule-based goal for the day, chosen from the user's weakest metric.
Daily Target	A user-set daily goal for steps, active minutes, water, or sleep.
Fitness Goal	A longer-term goal with a target value and a deadline.
Hash and salt	A hash is a one-way scramble of a password; a salt is random data added before hashing so equal passwords give different hashes.
Logging Streak	Consecutive days with at least one record of any type.
MET	Metabolic Equivalent of Task: the energy cost of an activity relative to rest, used to estimate calories.
Lockout	A 5-minute block on login after 5 consecutive failed attempts for a username.
Parameterized query	An SQL statement that passes user input separately from the command, preventing SQL injection.
Quick-add	A button that logs a preset amount of water in one click.
Reminder	A user-defined popup shown at a chosen time while the application is open.
Session	The period from login until logout or closing the application.
SQL injection	An attack that inserts SQL commands through input fields.
Wellness Score	A 0 to 100 daily indicator: steps 30 points, active minutes 25, water 20, sleep 25, each proportional to the share of target reached.